

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit in each work box show the operation(s) that you entered into the calculator and the result.

1. Aurora plays the trombone in her high school band. The table shows her practice times for one week.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Minutes	20	22	18	24	22	48	56

- a. Determine the range.

Range	38
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- b. Determine the mode.

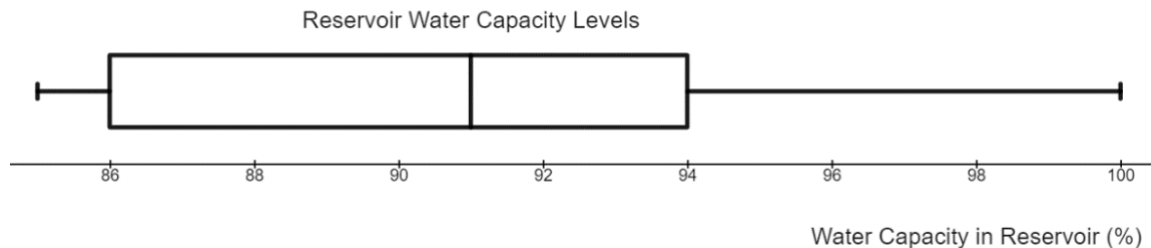
Mode	22
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- c. Determine the mean.

work	
Mean	30

2. In many western US states, reservoirs are used to collect and hold water that drains from melting snow or is runoff from rain. The capacity of the reservoir is written as a percentage. The data of the reservoir water capacity as a percentage for some water reservoirs in Wyoming for 1998 are shown in a list and a box plot.

{94%, 91%, 93%, 88%, 86%, 100%, 91%, 85%, 86%, 94%}



- a. Determine the smallest capacity for the reservoirs listed.

Smallest Capacity(%)	85
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- b. Determine the mean for the reservoirs listed.

work	
Mean	90.8

3. A local restaurant offers different items for dinner on its menu. The price of some items from the menu are shown.

{\$12.97, \$13.50, \$15.50, \$17.00, \$19.95, \$19.95, \$19.95, \$21.50}

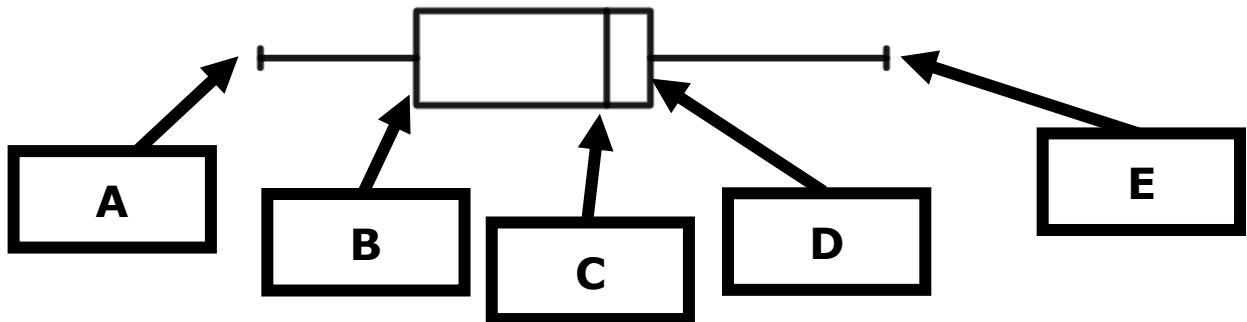
Determine and interpret the mode.

Mode	19.95
Interpretation of the Mode	There are more items on the menu that cost \$19.95 than any other price.

4. The price per box of oranges for the years 2007-2020 is shown.

{ \$6.61, \$5.77, \$6.96, \$8.41, \$9.92, \$7.43, \$9.63, \$9.39, \$9.90, \$11.82, \$12.44, \$11.41, \$9.27, \$9.53 }

A box plot of the data with some missing labels is shown.



a. Complete the statements.

The values between the box labeled

- ☐ **A**
- ☒ **B**
- ☐ **C**
- ☐ **D**
- ☐ **E**

and the box labeled

- ☐ **A**
- ☐ **B**
- ☐ **C**
- ☒ **D**
- ☐ **E**

represent the middle 50% of the data. The percentage of data between

the box labeled **C** and the box labeled **D** is

- ☐ less than
- ☐ more than
- ☒ the same as

the

percentage of data between the box labeled **D** and the box labeled **E**.

b. Determine the value that represents box **C**.

work	
Box C	9.46

c. Determine the range.

Range	6.67
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5. The table shows the question five different students wrote.

Student	Question
Erika	How many students are in the cafeteria today?
Jamal	How many times do students get their hair cut each year?
Nina	What is the average height of the sixth graders at our school?
Robert	What is your favorite club at the school?
Tiana	What is for lunch today?

- a. Complete the statement.

- ☐ Erika
- ☐ Jamal
- ☐ Nina
- ☐ Robert
- ☒ Tiana

did not write a statistical question because

the question

- ☐ is not numerical.
- ☐ is not categorical.
- ☒ does not have variability.

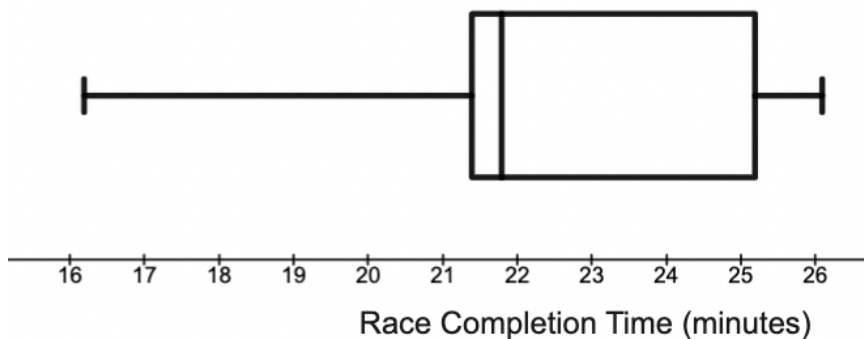
Student could also correctly complete the statement saying that Erika did not write a statistical question.

- b. Rewrite the question chosen in **part a** so that it is a statistical question.

Statistical Question	Answers will vary. Examples: • How long does it take students to walk to school? • What is the average distance students live from school? • What is the average amount of money students spend on lunch?
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6. A box plot representing the completion times, in minutes, of the 14 fastest runners in the local 5K race is shown.

5K Race Completion Results



	Value
Minimum	16.2
Lower Quartile	21.4
Median	21.8
Upper Quartile	25.2
Maximum	26.1

Complete the statements.

The fastest runner completed the race in

- ☒ 16.2
- ☐ 21.8
- ☐ 26.1

minutes.

Approximately 50% of the fastest runners completed the race between

- ☐ 16.2 and 21.4
- ☐ 16.2 and 25.2
- ☐ 16.2 and 26.1
- ☒ 21.4 and 25.2
- ☐ 21.4 and 26.1

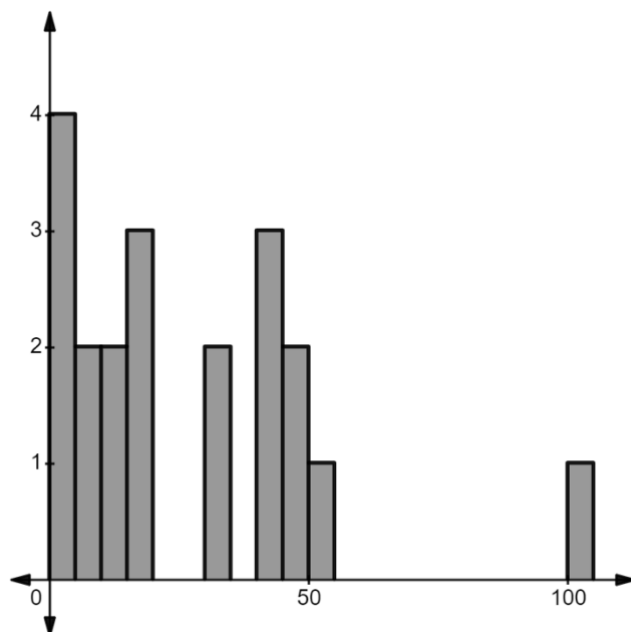
minutes. The interquartile range is

- ☐ 3.4
- ☒ 3.8
- ☐ 5.2
- ☐ 9.9

minutes.

7. Sixth graders were asked for the number of text messages they received on a particular day. A histogram representing the number of text messages a sample of 20 sixth graders received in one day is shown.

Number of Text Messages			
0	7	3	46
10	34	17	12
32	0	51	42
40	15	101	2
42	47	8	18



a. Complete the statements.

The shape of the distribution of received text messages can be best

described as

- ☒ skewed,
- ☐ uniform,
- ☐ symmetric,

with a

- ☐ gap
- ☐ peak
- ☒ cluster

between 40 and

55 text messages. This distribution has a

- ☒ gap
- ☐ peak
- ☐ cluster

between

20 and 30 text messages and an outlier between

- ☐ 30 and 35
- ☐ 55 and 100
- ☒ 100 and 105

text messages.

b. Determine the median.

work	
Median	17.5